Hints: Linear algebra – 10. Matrix norms and localisation of the spectrum

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Let $Ax = \lambda x$ and choose i with $|x_i|$ maximal. The i-th equation gives $|\lambda - a_{ii}|\,|x_i| \leqslant \sum_{j\neq i} |a_{ij}|\,|x_j|$.

Hint 2. Components: consider $A(t) = D + t(A - D)$, $t \in [0,1]$, with D the diagonal part. The discs grow continuously with t, and the eigenvalues depend continuously on t. An eigenvalue cannot jump from one component to a disjoint one, so the count in a component stays as at $t = 0$. Strict diagonal dominance means 0 lies in no disc.

Hint 3. For the example, take $A = \begin{pmatrix} 0 & 4 \\ 1 & 0 \end{pmatrix}$ with eigenvalues ± 2. Here row disc i intersected with column disc i is $\{|z| \leqslant 1\}$ for both i.

Problem 2

Hint 1. Courant–Fischer: if $\dim L = k$, then L meets the span of eigenvectors for $\lambda_k, \dots, \lambda_n$ (dimension $n - k + 1$). On that span the Rayleigh quotient is $\leqslant \lambda_k$. Equality is attained on the span of the first k eigenvectors. Also note the dual form $\lambda_k = \min_{\dim M = n-k+1} \max_{x\in M}$.

Hint 2. Cauchy: a principal $(n-1)\times(n-1)$ submatrix is the compression to a coordinate hyperplane H. Taking subspaces $L \subseteq H$ in the max-min (and in the min-max) gives $\lambda_k(A) \geqslant \mu_k \geqslant \lambda_{k+1}(A)$.

Hint 3. Weyl: use the min-max form with $M = M_A \cap M_B$. Here M_A is spanned by eigenvectors of A for $\lambda_i, \dots, \lambda_n$, and M_B by eigenvectors of B for $\lambda_j, \dots, \lambda_n$. Then $\dim M \geqslant n - (i + j - 1) + 1$, and on M the Rayleigh quotient of $A + B$ is at most $\lambda_i(A) + \lambda_j(B)$.

Problem 3

Hint 1. Order $\alpha_1 \geqslant \alpha_2$, $\beta_1 \geqslant \beta_2$, $\gamma_1 \geqslant \gamma_2$. Necessary conditions: $\gamma_1 + \gamma_2 = \alpha_1 + \alpha_2 + \beta_1 + \beta_2$ (trace), $\gamma_1 \leqslant \alpha_1 + \beta_1$, and by Weyl $\gamma_1 \geqslant \max(\alpha_1 + \beta_2, \alpha_2 + \beta_1)$.

Hint 2. Sufficiency: take $A = \operatorname{diag}(\alpha_1, \alpha_2)$ and $B = R_\theta \operatorname{diag}(\beta_1, \beta_2) R_\theta^T$ with R_θ a rotation. At $\theta = 0$ and $\theta = \pi/2$ you get the two ends of the interval; by continuity every γ_1 in between occurs. Answer: exactly the pairs with the right sum and $\gamma_1 \in [\max(\alpha_1 + \beta_2, \alpha_2 + \beta_1),\ \alpha_1 + \beta_1]$.

Problem 4

Hint 1. M is real symmetric, so all eigenvalues are real. Delete the first row and column: the remaining 9×9 matrix is the standard $\operatorname{tridiag}(-1, 2, -1)$. Its eigenvalues are $2 - 2\cos\frac{k\pi}{10} > 0$.

Hint 2. By Cauchy interlacing, $\lambda_1, \dots, \lambda_9$ of M are at least the smallest eigenvalue of the submatrix, so they are positive. Finally $\lambda_{10} \leqslant e_1^T M e_1 = -1 < 0$.

Problem 5

(a)

Hint 1. In terms of the real eigenvalues: $s = \sum \lambda_i^k = \|\lambda\|_k^k$ because k is even, and $t = \sum \lambda_i^{k+1}$ satisfies $|t| \leqslant \|\lambda\|_{k+1}^{k+1}$. Compare with the monotonicity $\|\lambda\|_{k+1} \leqslant \|\lambda\|_k$.

Hint 2. The hypothesis gives $|t| = s^{(k+1)/k}$, so $\|\lambda\|_{k+1} = \|\lambda\|_k$. Equality forces at most one non-zero eigenvalue. Then the characteristic polynomial is $x^{n-1}(x - \operatorname{tr} A)$; apply Cayley–Hamilton.

(b)

Hint 1. No. For $k = 1$ the condition is $(\sum \lambda_i)^2 = \sum \lambda_i^2$. Try $A = \operatorname{diag}(2, 2, -1)$ and compare A^3 with $\operatorname{tr}(A)A^2$.

Problem 6

Hint 1. Triangularise A, B simultaneously. The eigenvalues of AB are $\alpha_i\beta_i$, all on the unit circle.

Hint 2. A sum of 2018 unit complex numbers equals 2018 only if all of them are 1. So $\beta_i = \bar{\alpha}_i$ and $\operatorname{tr} B = \overline{\operatorname{tr} A} = \operatorname{tr} A$, since A is real.

Problem 7

Hint 1. $E + A^j$ is singular exactly when A has an eigenvalue λ with $\lambda^j = -1$. For which exponents j can one λ do this?

Hint 2. If $\lambda^j = -1$, then λ is a root of unity of order $2d$ with $d \mid j$ and j/d odd, so $v_2(j) = v_2(d)$. Hence one eigenvalue makes $E + A^j$ singular only for exponents j of one fixed 2-adic valuation.

Hint 3. The exponents $1, 2, 4, \ldots, 2^n$ have $n + 1$ different valuations, so $k = 2^n$ works. For $k = 2^n - 1$ take $A = \operatorname{diag}(e^{i\pi/2^a})_{a=0}^{n-1}$: every $j < 2^n$ has $a = v_2(j) \leqslant n - 1$, and $(e^{i\pi/2^a})^j = -1$ because $j/2^a$ is odd. Answer: $k = 2^n$.

Problem 8

Hint 1. $B = E - A$ commutes with A, so the identity holds for every eigenvalue pair a, $b = 1 - a$ (triangularise). Express everything through $p = ab$ using power sums with $a + b = 1$.

Hint 2. You get $(1 - 2p)(1 - 4p + 2p^2) = 1 - 5p + 5p^2$, i.e. $p(4p^2 - 5p + 1) = 0$. Invertibility excludes $p = 0$, so every $a_ib_i \in \{1, \frac{1}{4}\}$.

Hint 3. All combinations occur (diagonal A with entries $\frac{1}{2}$ or $\frac{1\pm i\sqrt{3}}{2}$). Answer: $\det(AB) \in \{4^{-m} : 0 \leqslant m \leqslant n\}$.

Problem 9

Hint 1. By homogeneity, $A(H_n) = H_n$ means $\|Ax\|_1 = \|x\|_1$ for all x. So A is an injective linear isometry, and it maps the unit ball $\operatorname{conv} H_n$ bijectively onto itself.

Hint 2. A linear bijection of a polytope onto itself maps vertices to vertices. Hence $Ae_i \in \{\pm e_j\}$, so A is a signed permutation matrix. There are $2^n n!$ of these.

Problem 10

Hint 1. Write $A = UD_aU^*$, $B = VD_bV^*$ and $W = U^*V$. Then $\|A - B\|_F^2 = \|D_aW - WD_b\|_F^2 = \sum_{i,j} |a_i - b_j|^2 |w_{ij}|^2$.

Hint 2. The matrix $(|w_{ij}|^2)$ is doubly stochastic. A linear function on the Birkhoff polytope attains its minimum at a vertex, i.e. at a permutation matrix (Birkhoff–von Neumann).

Hint 3. For the perturbation bound: $A\,\delta x = \delta b$ gives $\|\delta x\| \leqslant \|A^{-1}\|\,\|\delta b\|$, and $\|b\| \leqslant \|A\|\,\|x\|$. Multiply the two inequalities.